

PORTFOLIO PROJECT

BIS Analytics Pipeline

MS Fabric · dbt Core · Power BI · T-SQL

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Candidate — Senior Business Data Analyst

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Project Overview

Built a production-grade currency deposit analytics pipeline mirroring the BIS Banking Department tech stack — from synthetic Findur data to published Power BI report.

9

dbt models
across 4 layers

24

month-end
as-of dates

SDR 297bn

Avg active book

10

Power BI visuals
2 report pages

WHAT

- Currency deposit analytics
- Star schema (1 fact + 4 dims)
- Maturity ladder mart
- Fiscal year Apr → Mar (BIS convention)

WHY

- Demonstrates BIS-stack readiness
- Domain: SDR, FIXBIS, MTI, Sight, Notice
- Business logic from Annual Report
- Data validated against p.105-106

HOW

- Python → Fabric OneLake → dbt → Power BI
- 9 SQL models, all materialised as views
- Azure CLI auth → Fabric SQL Endpoint
- dbt run PASS=9 / Power BI refresh OK

BIS Domain Knowledge

All business logic validated against BIS Annual Report 2024/25

SDR — Unit of Account

p.103

Special Drawing Right — IMF basket of 5 currencies:

USD · EUR · GBP · JPY · CNY

BIS reports all figures in SDR — pipeline converts every deal to SDR via monthly IMF rate feed.

Deposit Products (Findur codes)

p.105

FIXBIS_DEP	Fixed-rate deposit — fixed term & rate. Most common.
MTI_DEP	Medium-term instrument — 1–10yr tenor.
SIGHT_DEP	On-demand / overnight. No maturity date.
NOTICE_DEP	Redeemable after advance notice period.

BIS Fiscal Year — April to March

p.101

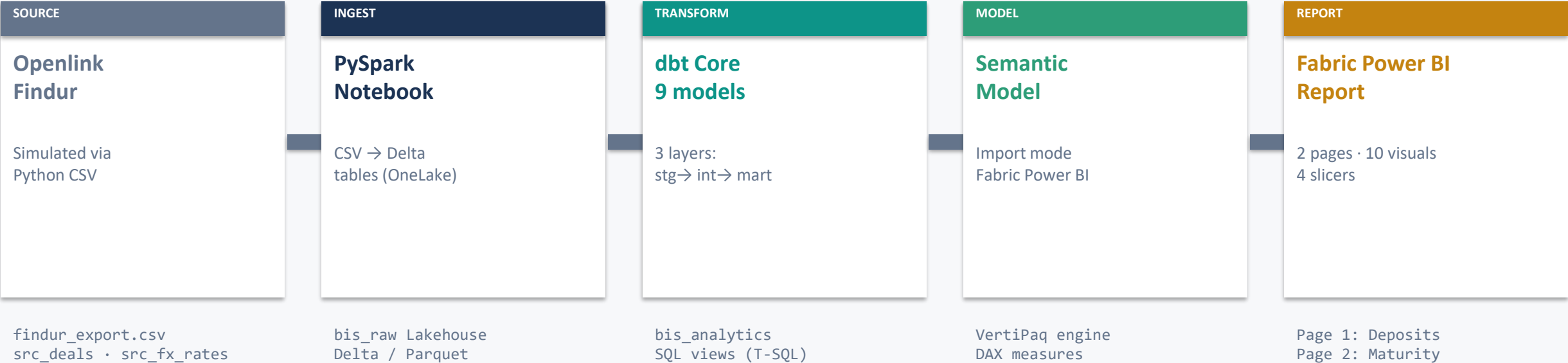
FY2023-24	1 Apr 2023 → 31 Mar 2024
FY2024-25	1 Apr 2024 → 31 Mar 2025 ← current
FQ1	April · May · June
FQ2–FQ4	Jul–Sep · Oct–Dec · Jan–Mar

Geographic Distribution of Deposits

p.106

Asia-Pacific	61%	
Americas	9%	
Europe	8%	
Middle East	7%	
Africa	6%	
International institutions	9%	

Pipeline Architecture

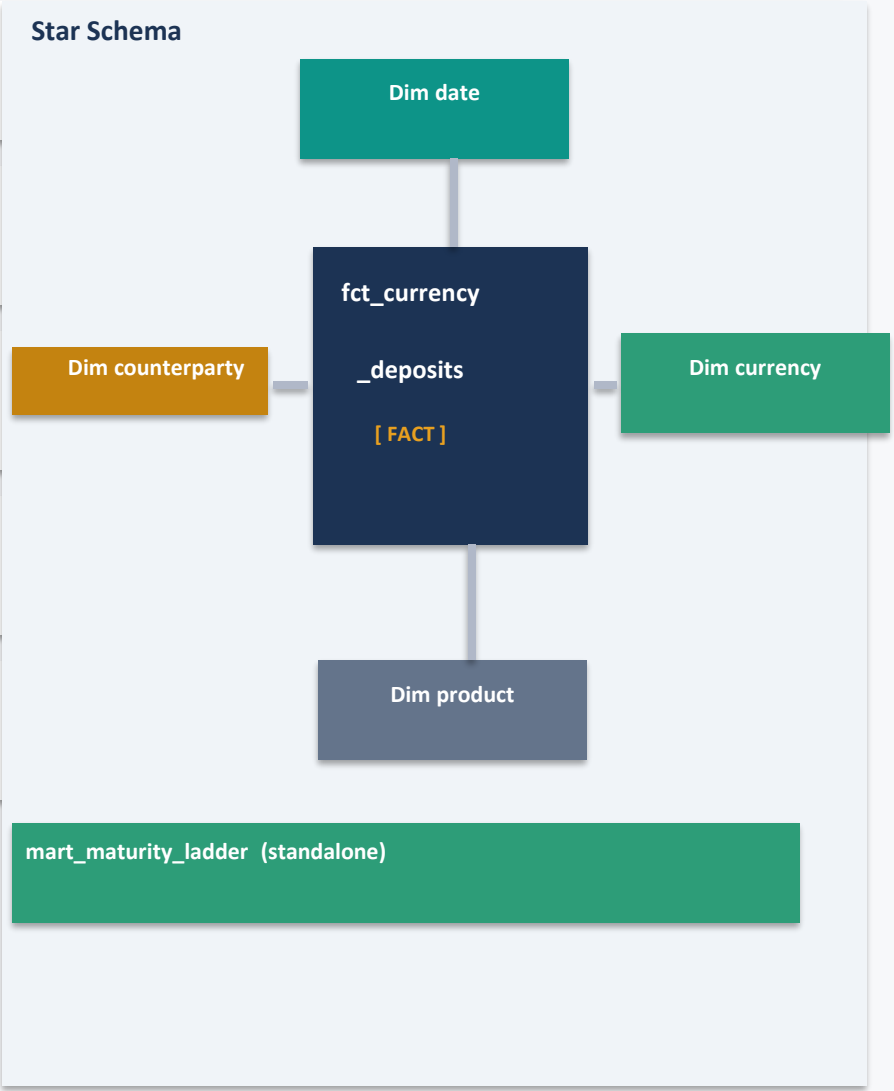
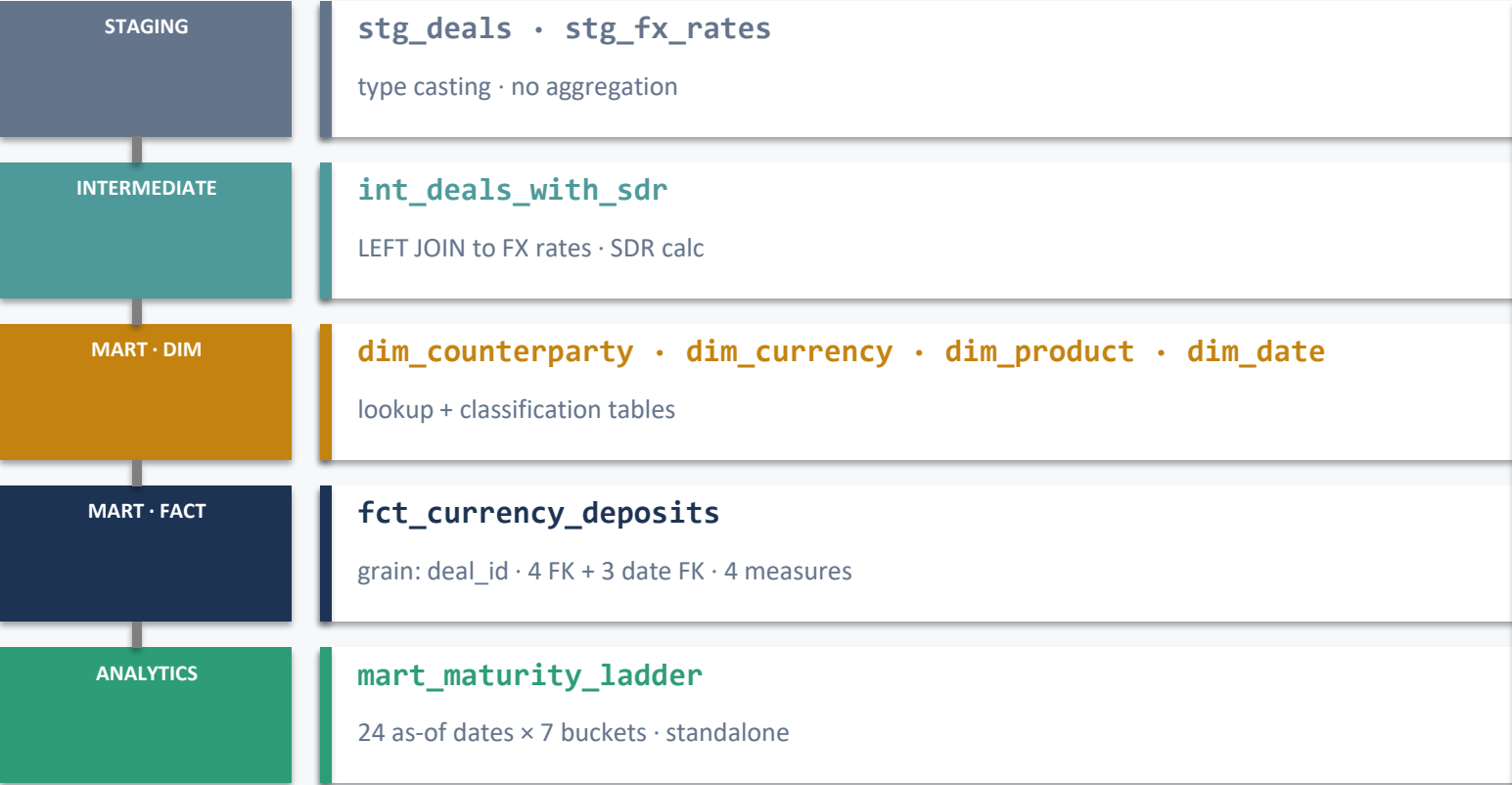


In BIS production:

Findur → Scheduled ETL pipeline (Spark / Cloudera CDS) · dbt Cloud or dbt Core · Lakehouse SQL endpoint (Delta / Parquet) · Power BI or Tableau with RLS · CI/CD pipeline (e.g. Azure DevOps)

Core skills transfer directly — star schema, dbt, SDR metrics apply on any Lakehouse platform.

Data Model — Star Schema



Maturity Ladder — Liquidity Risk Analytics

What is a maturity ladder?

A liquidity risk report showing how much of the deposit book matures in each time window at a specific point in time. Used daily by BIS treasury and risk teams.

7 maturity buckets:

1. Overnight	Sight accounts — redeemable any time
2. ≤ 1 Week	days_to_maturity ≤ 7
3. ≤ 1 Month	days_to_maturity ≤ 30
4. ≤ 3 Months	days_to_maturity ≤ 90
5. ≤ 6 Months	days_to_maturity ≤ 180
6. ≤ 1 Year	days_to_maturity ≤ 365
7. > 1 Year	days_to_maturity > 365

How it's built — hybrid approach

- 1 CROSS JOIN**
700 active deals × 24 month-ends
= 16,800 (deal, as-of date) pairs
- 2 days_to_maturity**
DATEDIFF(day, as_of_date, maturity_date) per pair
- 3 Bucket logic**
CASE WHEN in dbt SQL assigns each pair to 1 of 7 buckets
- 4 Aggregate**
COUNT + SUM per (as_of_date, bucket) → 71 rows output
- 5 Power BI slicer**
User picks as_of_date → chart shows that snapshot

✓ 71 rows · 24 dates × up to 7 buckets · dbt run PASS=1

Power BI Report — Deliverables

Page 1 — Deposit Book Overview

Stacked bar chart

Total SDR by Region — 6 regions, matches Annual Report Graph 4.3

Grouped bar chart

Deposits by Fiscal Year — FY2023-24 vs FY2024-25

Slicer (dropdown)

Fiscal Year filter

Slicer (dropdown)

Currency filter (10 currencies)

KPI cards

Total SDR · Deal Count · Avg Rate

Page 2 — Counterparty & Maturity

Table visual

Top counterparties: cpty_name · region · total SDR (Bn)

Bar chart

Maturity ladder — 7 buckets, sorted by time horizon

Slicer (dropdown)

as_of_date — 24 month-ends Apr 2023 → Mar 2025

Slicer (list)

Region filter

Interaction fix

FY slicer does not filter FY comparison chart (self-filter fix)

Tech Stack — Project vs BIS Production

Layer	This Project	BIS Production
Source system	Python-generated CSV (Findur simulation)	Openlink Findur (live extract)
Raw ingestion	PySpark notebook (CSV → Fabric Lakehouse)	Scheduled ETL pipeline (Spark / Cloudera CDS)
Transformation	dbt Core (CLI) Fabric SQL Analytics Endpoint	dbt Core or dbt Cloud (Databricks / Snowflake)
Semantic model	Fabric Power BI Import mode Fabric SQL Endpoint	Power BI or Tableau (direct connection)
Reporting	Fabric Power BI (workspace publish)	Power BI Service RLS · certified dataset
Auth	Azure CLI interactive (az login)	Service Principal Managed Identity
Orchestration	Manual CLI (dbt run)	CI/CD pipeline (e.g. Azure DevOps)
Architecture maps to modern data Lakehouse principles — dbt, SQL views, star schema, BI layer.		





Let's connect

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Role: Business Data Analyst

Stack: MS Fabric · dbt Core · Power BI · T-SQL

Project: BIS_Project_2026 — full pipeline available in the Git Hub repository:
https://github.com/gny-analytics/BIS_Analytics_Pipeline/tree/main
